

Hana Kouiriti

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EDUCATION

Northwestern Polytechnical University

Xi'an, China

B.Sc. in Computer Science

Expected Graduation: July 2027

- **CGPA:** 3.73/4.0

Intended Graduate Study: M.Sc. in Artificial Intelligence

Research Experience

Undergraduate Research Assistant for Prof. Zhaowen Liu

Northwestern Polytechnical University | China | 2026-Present

- Developing a hybrid system combining Large Language Models and Knowledge Graphs for mental health analysis.
- Designing pipelines to integrate unstructured clinical text with structured disorder and treatment knowledge.

Undergraduate Research Assistant for Prof. Zhang Lu

Northwestern Polytechnical University | China | 2025

- Trained transformer-based models on 20K+ power traces for side-channel analysis (SCAAM).
- Improved key byte identification accuracy by around 15% through architecture and hyperparameter optimization.

AWARDS & CERTIFICATES

First Prize: International Computer Science Competition (ICSC) 2025

Seniors Category

- Achieved highest nationwide score and Gold Honor, placing in the top 1% internationally for algorithmic reasoning and problem-solving.

First Prize: NPU Presidential Scholarship

- Awarded to the top 10% of students for outstanding academic performance at Northwestern Polytechnical University.

2nd Prize: AI Application Innovation Competition 2025

Team Leader, PearlAI Project

- Achieved second place in an AI innovation competition for developing PearlAI, a university assistant designed to support international students using large language models.

Star of Innovation in NPU for 2024-2025 Academic

- Recognized for innovation, problem-solving, and applying technology to meaningful academic and practical challenges.

VISION Global Competence Association – Activity Participation

Certificate *Northwestern Polytechnical University*

- Acknowledged for actively assisting in the organization and execution of intercultural events and student activities that promoted global competence and engagement.

Academic Excellence Certificates – Middle & High School

Alal El Fassi - Al Irfan High School

- Recognized for consistently ranking among the top students and graduating first in my major.

PROJECT EXPERIENCE

AI-Powered Drone Data Visualization

Northwestern Polytechnical University (Course Project) | China | 2025

- Built a Unity-based 3D drone swarm simulator with AI-driven task planning and a conversational agent for natural language drone control.
- Implemented multi-drone coordination, interactive 3D paths, signal/battery visualization, and multi-mode exploration (task planning, individual drones, comprehensive view).

AI Model Development: Heart Disease Prediction

Personal Project | 2024

- Trained a neural network model on a dataset of 10,000+ patient records using Python and TensorFlow.
- Achieved 92% prediction accuracy (outperforming baseline models by 15%) via feature engineering and hyperparameter tuning.
- Reduced false positives by 20% through stratified sampling and Synthetic Minority Over-sampling Technique (SMOTE) for class imbalance mitigation.

AI Assistant for University Students (PearlAi)

University Project | China | 2024

- Fine-tuned Qwen-2.5-32B on 1,500+ FAQs to support international students.
- Deployed chatbot on Alibaba Cloud server with 3,000+ monthly active users with <2s response time.
- Integrated retrieval-augmented search (RAG) for accurate and multilingual query handling.
- Project featured in official NPU News for pioneering AI-driven international student management.

Algorithm Visualization Platform (GitHub)

Course Project for Algorithms Design & Analysis | China | 2024

- Co-developed an interactive GitHub repository visualizing 5 core algorithms (sorting, searching, graph, DP, greedy) using Python.
- Implemented step-by-step animations and complexity analysis to improve conceptual understanding.

Competitive Programming & Open-Source

LeetCode, GitHub | 2023-Present

- Solved 400+ problems, specializing in dynamic programming and graph algorithms.
- Contributed to 6+ open-source projects, including a VS Code extension (BlackT codestyle) with 5,000+ downloads, featured in Harvard CS50's Gallery of Final Projects.

Game Development: Breakout & ShellStorm

Personal Projects | 2023

- Developed Breakout with 30+ levels and dynamic difficulty, achieving 60 Frames Per Second (FPS) on low-end hardware using OpenGL.
- Designed ShellStorm with 10+ enemy wave types; optimized collision detection to handle 1,000+ entities with <5ms latency.

ACTIVITIES & LEADERSHIP

ICPC Asia Xi'an Regional Contest | Technical Volunteer | 2025

- Contributed 90+ hours supporting contest systems, assisting teams with environment setup, and ensuring smooth technical operations.

BlackPearl | Co-founder & AI Lead | 2024–Present

- Led a team of 5 developers to build a university AI assistant, adopted by The International School of NPU (about 5000 users).

Recycling & Sustainability Fair | Participant | 2023

- Awarded a Certificate of Appreciation for creating a recycling project with a database and a Flask app to track and visualize waste reduction.

SKILLS

- **AI/ML:** Deep Learning (TensorFlow, PyTorch, Keras), Machine Learning (Scikit-learn, Pandas, NumPy), Natural Language Processing (transformers, embeddings, Large Language Model (LLM) fine-tuning), Retrieval-Augmented Generation (RAG), LangChain, Vector Databases (FAISS), HuggingFace.
- **Programming:** Python (primary), C/C++, Java, MATLAB, SQL.
- **Software & Frameworks:** Flask, FastAPI, Docker, Git, Linux/Ubuntu, Alibaba Cloud Deployment, Google Colab, Pygame.
- **Web Technologies:** JavaScript, HTML, CSS.
- **Languages:** Arabic (Native), English (Fluent), French (Fluent), Chinese (HSK4)